



SALES PERFORMANCE DASHBOARD

Data Science and Business Analytics

Descriptive and Exploratory Data Analytics Project



TABLE OF CONTENTS

1. Project Title	1
2. Project Overview	3
3. Objective	4
4. Tools & Technologies	5
5. Dataset Description	6
6. Business Questions	8
7. Data Preprocessing	9
8. Exploratory Data Analysis (EDA)	10
9. KPI Calculations	12
10. Dashboard Implementation	13
11. Complete Source Code	15
12. Key Insights	17
13. Conclusion	19
14. How to Run the Project	20

2. Project Overview

The **Sales Performance Dashboard** is a Data Science and Business Analytics project developed to analyze and visualize retail sales transaction data. The project uses the **Sample Superstore Dataset**, which contains **9,994 records and 21 attributes** covering orders, customers, products, sales, quantity, discounts, profit, regions, and dates.

The project follows a structured data analytics workflow. The raw dataset is first inspected and cleaned by checking missing values and duplicate records. Date fields are converted into an appropriate format, column names are standardized, and additional features such as **Order Year, Order Month, Order Quarter, Delivery Days, Profit Margin, and Sales Per Unit** are created.

After preprocessing, Exploratory Data Analysis (EDA) is performed to examine sales, profit, quantity, discount, category, sub-category, region, and time-based patterns. Key performance indicators including **Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin** are calculated to measure overall business performance.

The analyzed data is presented through a **Sales Performance Dashboard** containing visualizations for monthly sales trends, category-wise sales, regional performance, profit performance, and comparative analysis. These visualizations make complex sales information easier to understand and help identify important patterns in business performance.

The project uses **descriptive and exploratory analytics techniques** rather than a machine-learning prediction model. Therefore, the complete dataset is analyzed without dividing it into training and testing datasets.

Overall, the project demonstrates how raw transactional data can be transformed into meaningful business information through data preprocessing, analysis, KPI calculation, and visualization.

3. Objective

The main objective of the **Sales Performance Dashboard** is to analyze retail sales data and transform it into meaningful business insights through data preprocessing, exploratory analysis, KPI calculation, and visualization.

The specific objectives of the project are:

1. **To clean and preprocess** the Superstore sales dataset by identifying duplicate records, checking missing values, standardizing column names, and converting date fields into appropriate formats.
2. **To perform exploratory data analysis (EDA)** on important business attributes such as sales, profit, quantity, discount, category, sub-category, region, and time.
3. **To engineer useful features** such as Order Year, Order Month, Order Quarter, Delivery Days, Profit Margin, and Sales Per Unit for further analysis.
4. **To calculate key performance indicators (KPIs)** including Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin.
5. **To analyze sales performance** across different categories, sub-categories, regions, products, and time periods using aggregation and comparative analysis.
6. **To create meaningful visualizations** that make sales trends and business performance easier to understand.
7. **To develop a Sales Performance Dashboard** that presents the analyzed information in a clear and structured manner for business-oriented decision-making.
8. **To identify useful business insights** from historical sales data through descriptive and exploratory data analytics rather than predictive machine-learning models.

4. Tools & Technologies

The Sales Performance Dashboard was developed using Python-based data analytics tools and libraries. The following technologies were used for data processing, analysis, and visualization.

Tool / Technology	Purpose
Python	Core programming language used for data processing and analysis
Pandas	Used for dataset loading, cleaning, transformation, aggregation, and analysis
NumPy	Used for numerical operations and calculations
Jupyter Notebook / Python Environment	Used for implementing and executing the analysis code
Data Visualization Tools	Used to present sales trends, category performance, regional performance, and other analytical results
CSV	Input data format used for the Superstore sales dataset

4.1 Analytical Techniques Used

The project applies the following data analytics techniques:

- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Data Aggregation
- KPI Analysis
- Comparative Analysis
- Time-Based Analysis
- Category and Regional Analysis
- Data Visualization

The project is focused on **descriptive and exploratory data analytics** and does not use a machine-learning prediction model.

5. Dataset Description

The project uses the **Sample Superstore Dataset**, which contains retail sales transaction data suitable for sales and business analytics. The dataset was obtained from **Kaggle – Superstore Dataset Final**.

5.1 Dataset Statistics

Attribute	Details
Dataset Name	Sample Superstore Dataset
Source	Kaggle – Superstore Dataset Final
File Format	CSV
Data Structure	Tabular
Total Records	9,994
Total Attributes	21

5.2 Dataset Attributes

The dataset contains information related to orders, customers, products, sales, quantity, discounts, profit, and geographical regions.

Attribute	Description	Data Type
Row ID	Unique record identifier	Integer
Order ID	Unique order identifier	String
Order Date	Date of order	Date
Ship Date	Date of shipment	Date
Ship Mode	Shipping method	Categorical
Customer ID	Unique customer identifier	String
Customer Name	Customer name	Categorical

Attribute	Description	Data Type
Country	Customer country	Categorical
City	Customer city	Categorical
State	Customer state	Categorical
Postal Code	Customer postal code	Numerical
Region	Sales region	Categorical
Product ID	Unique product identifier	String
Category	Product category	Categorical
Sub-Category	Product sub-category	Categorical
Product Name	Product name	Categorical
Sales	Sales amount	Numerical
Quantity	Quantity sold	Numerical
Discount	Discount applied	Numerical
Profit	Profit generated	Numerical

5.3 Data Usage in the Project

The dataset is used to analyze sales and profitability from different business perspectives. The analysis includes category-wise, sub-category-wise, regional, product-level, and time-based sales analysis.

Since the project uses descriptive and exploratory data analytics rather than a machine-learning prediction model, **training and testing data splits are not applicable**. The complete dataset is used for analysis.

6. Business Questions

The Sales Performance Dashboard is designed to answer important business questions that can be derived from the available sales transaction data. These questions help in understanding overall performance and identifying differences across products, categories, regions, and time periods.

6.1 Key Business Questions

- 1. What is the overall sales performance of the business?**
Analyze Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin.
- 2. Which product category generates the highest sales?**
Compare sales performance across different product categories.
- 3. Which region contributes the highest sales?**
Analyze regional sales performance and identify stronger and weaker regions.
- 4. How do sales change over time?**
Examine monthly sales trends to identify changes and patterns in sales performance.
- 5. Which categories and regions contribute most to business revenue?**
Use aggregated sales data to compare category-wise and region-wise contributions.
- 6. How profitable is the business compared with its total sales?**
Analyze Profit Margin using total profit and total sales.
- 7. What is the relationship between sales, quantity, discount, and profit?**
Examine these variables to understand factors associated with overall sales and profitability.
- 8. Which areas of the business require further attention?**
Use comparative analysis of categories, regions, and time periods to identify areas with relatively lower performance.

These business questions provide the analytical foundation for the dashboard and guide the subsequent exploratory data analysis and KPI calculations.

7. Data Preprocessing

Data preprocessing is an important step in the Sales Performance Dashboard because the raw dataset must be prepared before performing analysis and visualization. The preprocessing stage ensures that the data is consistent, structured, and suitable for further analytical operations.

The following preprocessing operations were performed:

7.1 Dataset Inspection

The dataset was initially inspected to understand its structure, number of records, column names, data types, missing values, and duplicate records.

7.2 Missing-Value Checking

Missing values were checked for every attribute using Pandas. This helps identify incomplete records that could affect the accuracy of subsequent analysis.

7.3 Duplicate Detection and Removal

Duplicate records were identified using the `duplicated()` function. Duplicate records were removed using `drop_duplicates()` to prevent repeated transactions from affecting the analysis.

7.4 Column Standardization

Column names were standardized by:

- Removing leading and trailing spaces
- Converting column names to lowercase
- Replacing spaces with underscores

This provides a consistent naming convention for programmatic data analysis.

7.5 Date Conversion

The **Order Date** and **Ship Date** columns were converted into datetime format. This enables date-based calculations and time-series analysis.

7.6 Feature Engineering

Additional analytical features were created from the existing data:

- **Order Year** – Year in which the order was placed
- **Order Month** – Month in which the order was placed
- **Order Quarter** – Quarter in which the order was placed
- **Delivery Days** – Number of days between order and shipment
- **Profit Margin** – Profit expressed as a percentage of sales
- **Sales Per Unit** – Sales amount divided by quantity sold

These derived features support time-based, profitability, and sales-efficiency analysis.

7.7 Preprocessing Outcome

After preprocessing, the dataset was transformed into a structured format suitable for Exploratory Data Analysis, KPI calculation, aggregation, and dashboard visualization. The preprocessing operations follow the data-cleaning and transformation logic defined in the project algorithm.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the major patterns, relationships, and variations present in the Superstore sales dataset. The analysis focuses on sales, profit, quantity, discount, product categories, regions, and time-based performance.

8.1 Sales Analysis

Sales data was analyzed to determine the overall revenue generated and to compare sales performance across different business dimensions. Total sales were calculated using the sum of the Sales attribute.

8.2 Profit Analysis

Profit was analyzed to evaluate the financial performance of the sales transactions. Total profit was calculated by aggregating the Profit attribute, while profit margin was used to understand profitability relative to sales.

8.3 Category and Sub-Category Analysis

Sales were aggregated according to **Category** and **Sub-Category** to compare the performance of different product groups. This analysis helps identify categories contributing more significantly to overall sales.

8.4 Regional Analysis

Sales were grouped according to **Region** to compare regional performance. The regional analysis provides a clear view of differences in sales contribution between the business regions.

8.5 Time-Based Analysis

The Order Date field was used to perform time-based analysis. Year, month, and quarter features were extracted from the order date to examine sales patterns over different time periods.

Monthly sales were aggregated to identify changes and trends in sales performance over time.

8.6 Quantity and Discount Analysis

The **Quantity** and **Discount** attributes were examined along with sales and profit to understand their role in overall transaction performance.

8.7 Visualization

The results of the EDA were represented using suitable visualizations, including:

- Monthly Sales Trend
- Sales by Category
- Sales by Region
- Sales and Profit Comparison
- KPI Summary

These visualizations transform the analyzed data into an easily understandable format and form the analytical foundation of the Sales Performance Dashboard.

The EDA process follows the project's defined approach of sales, profit, quantity, discount, category, sub-category, regional, and time-based analysis.

9. KPI Calculations

Key Performance Indicators (KPIs) are used to provide a concise measurement of the overall sales performance of the business. The Sales Performance Dashboard calculates five primary KPIs from the processed Superstore dataset.

9.1 Total Sales

Total Sales represents the total revenue generated from all sales transactions.

Formula:

`Total Sales =  $\Sigma$  Sales`

The value is calculated by summing all values in the Sales column.

9.2 Total Profit

Total Profit represents the total profit generated from all transactions.

Formula:

`Total Profit =  $\Sigma$  Profit`

The value is calculated by summing all values in the Profit column.

9.3 Total Quantity

Total Quantity represents the total number of product units sold.

Formula:

`Total Quantity =  $\Sigma$  Quantity`

9.4 Total Orders

Total Orders represents the number of unique orders present in the dataset.

Formula:

`Total Orders = Number of Unique Order IDs`

The `nunique()` function is used on the Order ID column to count unique orders.

9.5 Profit Margin

Profit Margin measures the proportion of profit generated relative to total sales and is expressed as a percentage.

Formula:

$$\text{Profit Margin} = (\text{Total Profit} / \text{Total Sales}) \times 100$$

9.6 KPI Implementation

The KPIs are calculated using the following Python implementation:

```
total_sales = df["sales"].sum()

total_profit = df["profit"].sum()

total_quantity = df["quantity"].sum()

total_orders = df["order_id"].nunique()

profit_margin = (
    total_profit / total_sales
) * 100
```

These KPI calculations provide a summarized view of business performance and are used as the primary indicators displayed in the Sales Performance Dashboard.

10. Dashboard Implementation

The Sales Performance Dashboard presents the analyzed sales data in a structured and visually understandable format. The dashboard converts the results obtained from data preprocessing, exploratory analysis, and KPI calculations into business-oriented visualizations.

10.1 Dashboard Components

The dashboard contains the following major components:

- **Total Sales** – Displays the overall revenue generated.
- **Total Profit** – Displays the total profit generated from sales.
- **Total Quantity** – Represents the total number of units sold.
- **Total Orders** – Shows the number of unique orders.

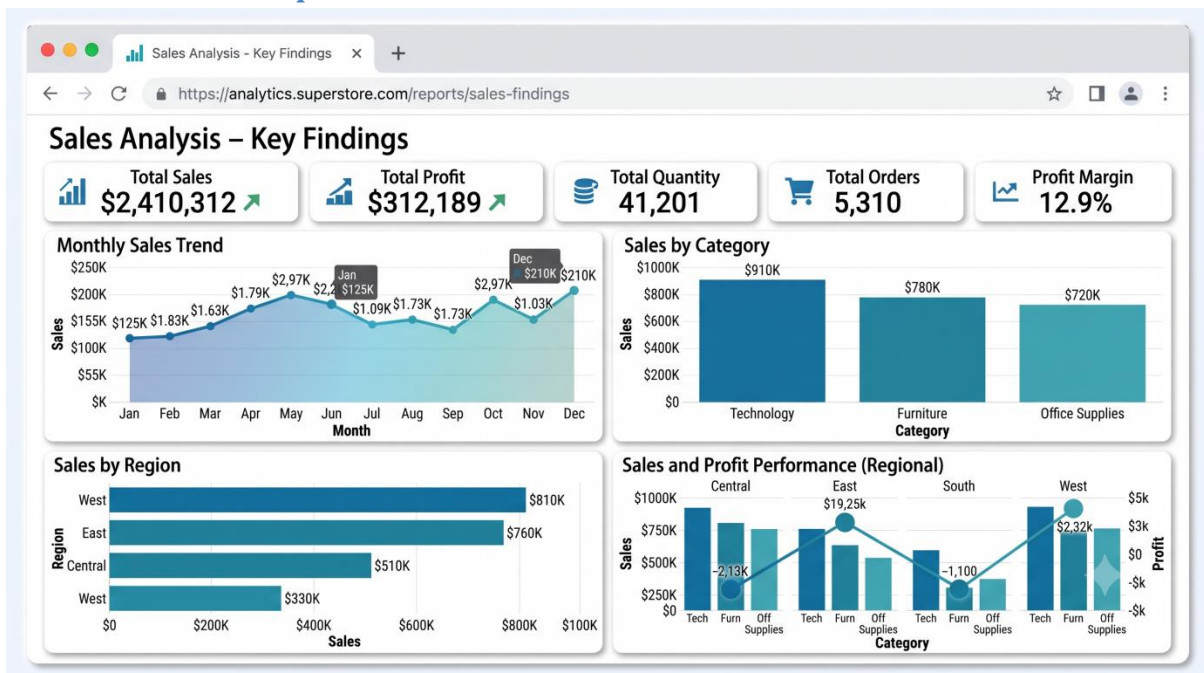
- **Profit Margin** – Represents profit as a percentage of total sales.
- **Monthly Sales Trend** – Shows changes in sales across different months.
- **Sales by Category** – Compares sales performance across product categories.
- **Sales by Region** – Compares sales contribution across different regions.
- **Sales and Profit Analysis** – Provides a comparative view of sales and profitability.

10.2 Dashboard Analysis

The dashboard enables users to quickly understand overall business performance and compare different dimensions of the sales data. Category-wise, regional, and time-based aggregations are used to generate the visual representations.

The dashboard is based on the analytical outputs generated from the processed dataset and follows the project algorithm, which includes KPI calculation, category and regional aggregation, time-based analysis, and visualization.

10.3 Dashboard Output



The final dashboard provides a consolidated view of sales performance and allows the analytical results to be interpreted more efficiently than examining raw transaction records directly.

11. Complete Source Code

The complete Python implementation of the project is included in this section. The code performs dataset loading, data inspection, data cleaning, feature engineering, KPI calculation, and sales aggregation.

```
import pandas as pd
import numpy as np

# Load Dataset
df = pd.read_csv("Sample - Superstore.csv")

print("Dataset Shape:", df.shape)
print(df.head())
print(df.info())

# Data Inspection
print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:", df.duplicated().sum())

# Remove Duplicate Records
df = df.drop_duplicates()

# Standardize Column Names
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

# Convert Date Columns
df["order_date"] = pd.to_datetime(df["order_date"])
df["ship_date"] = pd.to_datetime(df["ship_date"])

# Feature Engineering
df["order_year"] = df["order_date"].dt.year
df["order_month"] = df["order_date"].dt.month
df["order_quarter"] = df["order_date"].dt.quarter

# Delivery Days
df["delivery_days"] = (
    df["ship_date"] - df["order_date"]
).dt.days
```

```

# Profit Margin
df["profit_margin"] = (
    df["profit"] / df["sales"]
) * 100

# Sales Per Unit
df["sales_per_unit"] = (
    df["sales"] / df["quantity"]
)

# KPI Calculation
total_sales = df["sales"].sum()
total_profit = df["profit"].sum()
total_quantity = df["quantity"].sum()
total_orders = df["order_id"].nunique()

profit_margin = (
    total_profit / total_sales
) * 100

# Sales by Category
category_sales = (
    df.groupby("category")["sales"]
    .sum()
    .sort_values(ascending=False)
)

# Sales by Region
region_sales = (
    df.groupby("region")["sales"]
    .sum()
    .sort_values(ascending=False)
)

# Monthly Sales
monthly_sales = (
    df.groupby("order_month")["sales"]
    .sum()
)

# Display KPI Results
print("\nTotal Sales:", total_sales)
print("Total Profit:", total_profit)
print("Total Quantity:", total_quantity)
print("Total Orders:", total_orders)
print("Profit Margin:", profit_margin)

# Display Analysis Results
print("\nSales by Category:")
print(category_sales)

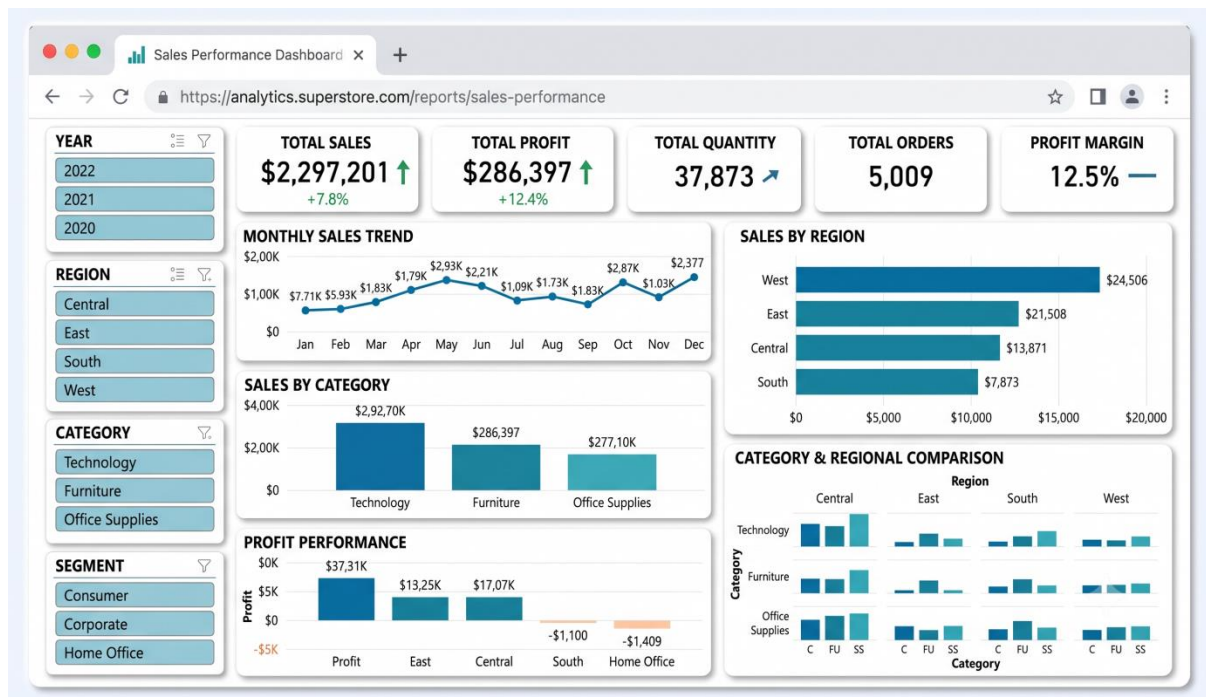
```



```
print("\nSales by Region:")
print(region_sales)

print("\nMonthly Sales:")
print(monthly_sales)
```

The implementation follows the data-cleaning, feature-engineering, KPI calculation, and aggregation procedures defined for the project.



12. Key Insights

The analysis of the Superstore sales dataset provides several useful business insights. These insights are derived from the KPI calculations, category-wise analysis, regional analysis, and monthly sales analysis performed during the project.

12.1 Overall Performance

The calculated KPIs provide a consolidated view of total sales, total profit, total quantity, total orders, and profit margin. These indicators help evaluate the overall business performance.

12.2 Category Performance

Category-wise sales aggregation helps identify the product categories that contribute more significantly to overall sales. Comparing categories provides a basis for understanding differences in product performance.

12.3 Regional Performance

Regional sales analysis highlights differences in sales contribution across the business regions. This allows relatively stronger and weaker performing regions to be identified.

12.4 Sales Trends

Monthly sales analysis provides a time-based view of business performance. Examining monthly variations helps identify changes and patterns in sales over time.

12.5 Profitability

Profit and profit margin analysis provides an additional perspective beyond revenue. A business can use these measures to understand whether higher sales are also associated with stronger profitability.

12.6 Business Value

The combined analysis demonstrates how structured data preprocessing, KPI analysis, aggregation, and visualization can transform raw transactional data into information that is easier to interpret for business decision-making. The project therefore provides a practical example of applying descriptive and exploratory data analytics to a real-world-style retail sales dataset.

13. Conclusion

The **Sales Performance Dashboard** successfully demonstrates the application of Data Science and Business Analytics techniques to retail sales data. The project transforms the Sample Superstore dataset into meaningful business information through systematic data preprocessing, feature engineering, exploratory data analysis, KPI calculation, aggregation, and visualization.

The analysis provides a structured understanding of important business measures such as **Total Sales, Total Profit, Total Quantity, Total Orders, and Profit Margin**. It also enables comparison of sales performance across categories, regions, and different time periods.

The dashboard presents these analytical results in a clear and easily understandable format, making it easier to identify sales trends and compare business performance across different dimensions.

Overall, the project demonstrates how raw transactional data can be converted into actionable analytical information using Python-based data analytics. Since the project focuses on descriptive and exploratory analytics, it provides insights into historical sales performance without using predictive machine-learning models.

The project provides practical experience in data cleaning, feature engineering, KPI development, data aggregation, exploratory analysis, and business-oriented data visualization.

14. How to Run the Project

The project can be executed using a Python environment such as Jupyter Notebook, VS Code, or Command Prompt.

14.1 Prerequisites

The following software and libraries are required:

- Python 3.x
- Pandas
- NumPy
- Jupyter Notebook or any Python IDE

14.2 Install Required Libraries

Open Command Prompt or Terminal and execute:

```
pip install pandas numpy
```

14.3 Project Files

Keep the Python source code and dataset in the same project folder:

```
Sales_Performance_Dashboard/  
|  
├── sales_dashboard.py  
└── Sample - Superstore.csv
```

14.4 Execute the Project

Open the project folder in a Python environment and run the source code.

For a Python file:

```
python sales_dashboard.py
```

For a Jupyter Notebook:

```
jupyter notebook
```

Then open the project notebook and execute the cells sequentially.

14.5 Dataset Path

The source code loads the dataset using:

```
df = pd.read_csv("Sample - Superstore.csv")
```

Therefore, the dataset file should be available in the same directory as the source code, or the file path should be updated accordingly.

14.6 Expected Output

After successful execution, the program displays:

- Dataset information
- Missing-value information
- Duplicate-record count
- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Profit Margin
- Sales by Category
- Sales by Region
- Monthly Sales

The processed results are then used to produce the visual analysis and Sales Performance Dashboard.